



Article title

Multi-Class Oil Spill Segmentation in SAR Images Using DeepLabV3+ with Progressive Optimization

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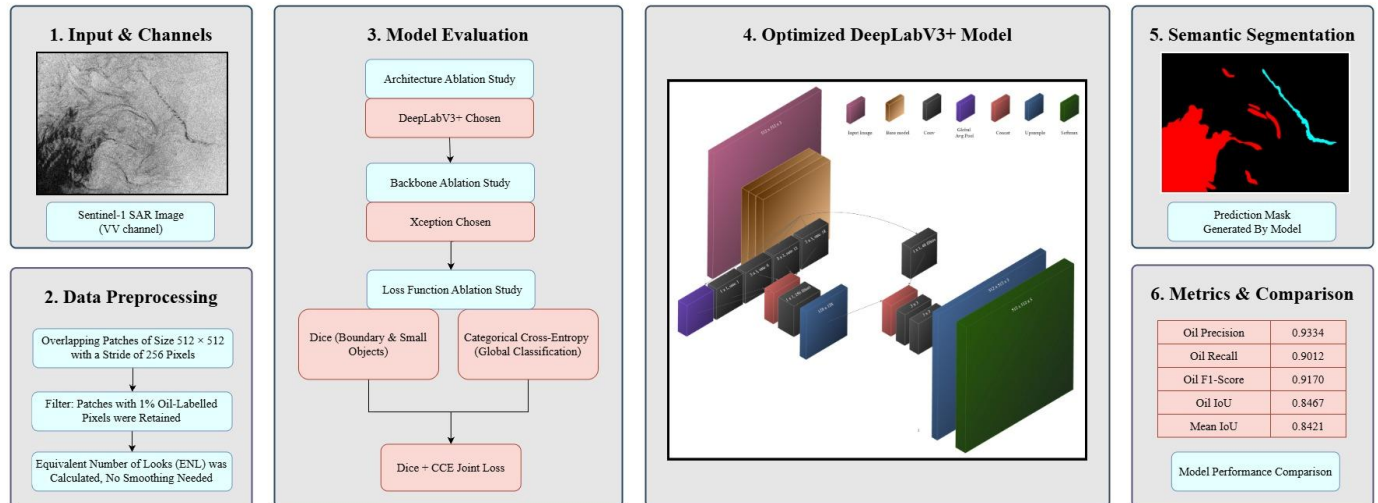
Abstract

Oil spills pose a serious threat to the marine ecosystem and require reliable large-scale monitoring. Synthetic Aperture Radar (SAR) imagery enables continuous oil spill monitoring under all weather and day-night supervision, but accurate segmentation remains a challenge due to speckle noise, class imbalance, and look-alike phenomena. This study develops an optimized deep learning framework for the segmentation of oil spills using Sentinel-1 SAR imagery. A DeepLabV3+ model is systematically optimized through backbone selection and progressive loss selection to enhance contextual feature learning. Experimental results demonstrate that the optimized DeepLabV3+ model achieves accurate results compared to a U-Net baseline.

- Formulating marine oil spill detection from SAR (Synthetic Aperture Radar) imagery as a multi-class semantic segmentation problem, addressing SAR-specific challenges.
- The use of advanced encoder-decoder architectures designed to decouple spatial and channel information for more precise boundary localization.
- A joint-loss approach that balances global class distribution to minimize false alarms.

The resulting model achieves an Oil Precision of 0.9334, an Oil Recall of 0.9012, an Oil F1-score of 0.9170, an Oil-Spill IoU of 0.8467, and a Mean IoU of 0.8421. These advancements provide a more accurate and computationally efficient solution for real-time marine surveillance and environmental protection.

Graphical abstract



Specifications table

Subject area	Computer Science
More specific subject area	Deep Learning, Object Detection, Image Segmentation
Name of your method	Semantic Segmentation, Vision Transformers, DeepLabV3+

Background

Around 71% of Earth's surface is covered by water, with oceans being the majority [1]. Oceans not only provide the habitat for diverse marine biology on earth, but also provide resources for human beings. However, it's constantly threatened by pollution. Statistically, 5.86 million tons of oil have leaked from seaborne vessels alone in the past 50 years [2]. A large amount of toxic compounds and heavy metals are present in crude oil, which rapidly enter the marine ecological cycle through the food chain and damage the health of the marine ecosystem once an oil spill occurs [3]. Therefore, timely and accurate detection of marine oil spills is important for effectively starting the oil-spill cleaning process and the protection of the marine environment. To manage and minimize oil spills, they should first be identified.

The advent of deep learning technology has been propelled by the accumulation of datasets, by the availability of large-scale datasets, and concurrent advancements in computing capabilities [4]. Satellite imaging can be an efficient tool for this purpose. Advancements in satellite remote sensing technology, Synthetic Aperture Radar (SAR) technology has proven to be an effective tool for the detection of spilled oil independent of sunlight and weather conditions. SAR systems operate 24/7, making them well-suited for continuous ocean monitoring [5]. In SAR imagery, the oil slicks

appear as dark patches caused by the low back-scatter response compared to the surrounding clean sea area [6]. Crude oils are typically brown to black when on the sea in thicker slicks ($>3 \mu\text{m}$). Fuel oils are black [7].

Despite the advantages of SAR technology, oil spill detection remains challenging. The dark patches in SAR imagery can also be caused by low-wind areas, biogenic films, or other lookalike phenomena, leading to false detection of oil [8]. In addition to this, SAR images are affected by speckle noise and complex sea surface conditions, further complicating the result [9]. As a result, solely relying on simple detection or handcrafted features is often insufficient, motivating the use of data-driven deep learning-based approaches [10]. This paper presents a comprehensive study of deep learning based semantic segmentation approaches for marine oil spill detection using SAR imagery, with a focus on evaluating model performance under challenges such as speckle noise and look-alikes.

Method details

Dataset

The SAR dataset used in this study was compiled as a benchmark dataset for oil spill segmentation using images from the Sentinel-1 satellites [10][11]. Originally, the images were a part of the European Space Agency's Copernicus Open Access Hub, while oil pollution events were verified through records provided by the European Maritime Safety Agency (EMSA) CleanSeaNet service [12]. As Fig. 1 shows, the dataset comprises single polarization channel (VV channel) imagery. All images are resized to a spatial resolution of 1250×650 pixels. Sentinel-1 satellites operate a C-band SAR sensor with a ground pixel spacing of approximately 10×10 m, enabling wide-area maritime monitoring [13]. Although dual polarization data (VV and VH) are available, only the VV polarization channel is utilized in this work, following the dataset's original configuration.

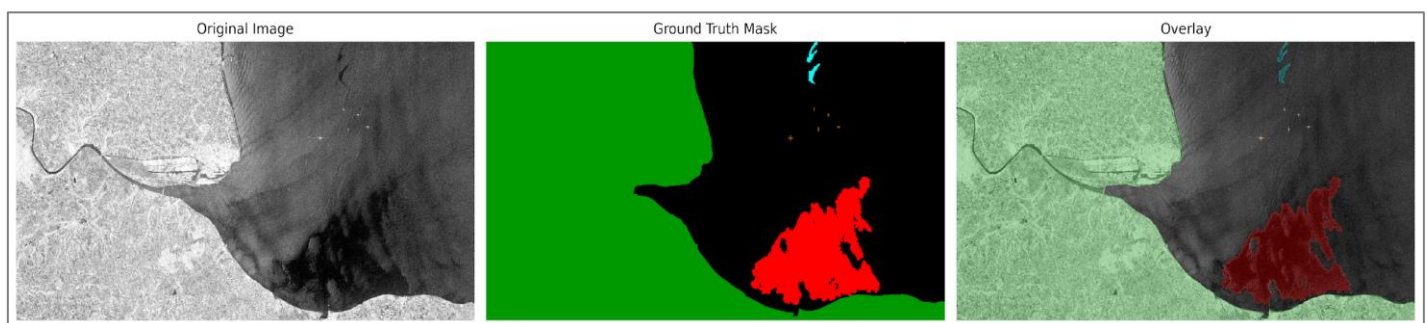


Fig. 1: Image from the Dataset with the Corresponding Mask Containing All 5 Classes.
Background: Black, Oil-Spill: Cyan, Look-Alike: Red, Ship: Brown, Land: Green.
Left: Input SAR Image, Middle: Ground Truth Mask, Right: Mask Overlayed on Input Image

The dataset contains 1112 SAR images, each of which has a corresponding segmentation mask. The masks are made out of 5 unique colours, each representing a class. The five semantic classes include background, oil spill, look-alike, ship, and land. The dataset is originally split into training and testing subsets in a 90-10 ratio. Training set containing 1002 images and the testing set containing 110. Due to the nature of oil spills, the dataset exhibits class imbalance with

the background (sea-surface) class dominating most of the dataset. Oil spills, ship and look-alikes occupy very small regions throughout. The creators of the dataset have also included multiple zoom levels to improve robustness against size differences and altitude.

Data Preprocessing

The original resolution of the images in the dataset is 1250×650 , which is computationally expensive to load up for training. Therefore, a patching strategy was implemented. The SAR images were 3-channel; all the 3 channels stored the exact same values, so 2 redundant channels were removed. Only while training the model, the inputs were replicated thrice to conform to the input shape of various backbones and architectures. For each patch, the image was first normalised. The mask from Fig. 1 (middle) underwent a mapping function where colours were mapped to actual class labels $\in \{0,1,2,3,4\}$.

The images and masks were converted from .jpg and .png formats to .npy (float32) and .npy (uint8), respectively. The .npy files for each image and mask contain the true array values, which load faster and also don't go through any implicit decoding transformations, thereby speeding up training as well. Although storage space is higher for .npy files as they aren't compressed, getting rid of the redundant channels vastly reduced the size of the dataset, allowing for even quicker loading.

Each image was split into overlapping patches of size 512×512 with a stride of 256 pixels to quicken training, but at the same time preserving spatial information. This also helps by increasing the number of training samples, with background (sea-surface) being the dominant class; multiple patches only contained sea-surface, worsening the class imbalance. To fix this, patches containing 1% oil-labelled pixels were retained, and the rest were discarded, the 1% figure was chosen because there were instances of oil spills occupying extremely small regions. As shown in Fig. 2, to assess the presence of speckle noise in the dataset, Equivalent Number of Looks (ENL) was calculated on 50×50 sea-surface patches across randomly sampled instances from the dataset [14].

$$ENL = \frac{\mu^2}{\sigma^2}$$

Here, μ is the mean, and σ is the standard deviation of intensity (pixel-values) in a homogenous patch. Higher ENL values suggest less speckle noise and vice versa.

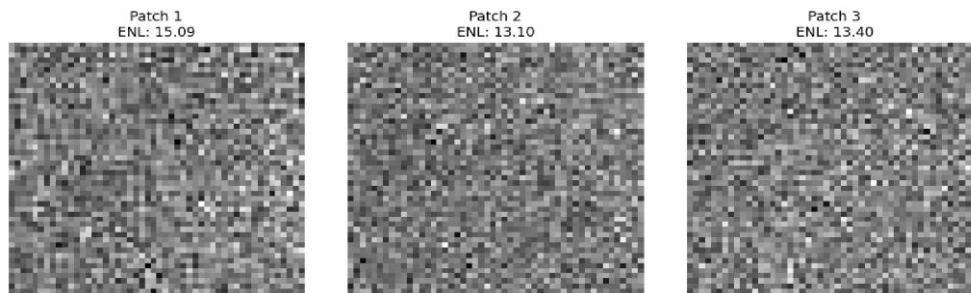


Fig. 2: Three Separate 50 x 50 patches of Sea-Surface class with ENL Values.

Data augmentation techniques, including horizontal and vertical flips, 90-degree rotation, and low-variance Gaussian noise, were introduced to improve model robustness [15]. These augmentations were chosen because they still end up producing realistic SAR images that could occur naturally in the dataset. The augmentations were exclusively applied to the training set only. The original dataset followed a 90-10 train-test split. The patch dataset generated was divided into 80% training, 10% validation, using the original training set. The 10% testing set was kept as is.

Model Architectures

1. U-Net: We used a U-Net architecture with skip connections and used ResNet-50 as the encoder [16]. From Fig. 3, the architecture has an encoder-decoder design with skip connections capturing both high-level and fine-grained spatial features. Encoder creates the latent representation using convolutional and pooling layers, while the decoder upsamples to help form the predicted mask using the spatial information stored in the bottleneck by transposed convolutions. Skip connections between the encoder and decoder parts are crucial in accurately keeping track of boundary details, thereby aiding accurate segmentation. A final 1×1 convolution with SoftMax activation produces the final segmentation mask, with each pixel getting labelled with a class. The ability of the architecture to preserve features at all levels and localisation is what makes it a good fit for semantic segmentation on SAR images.

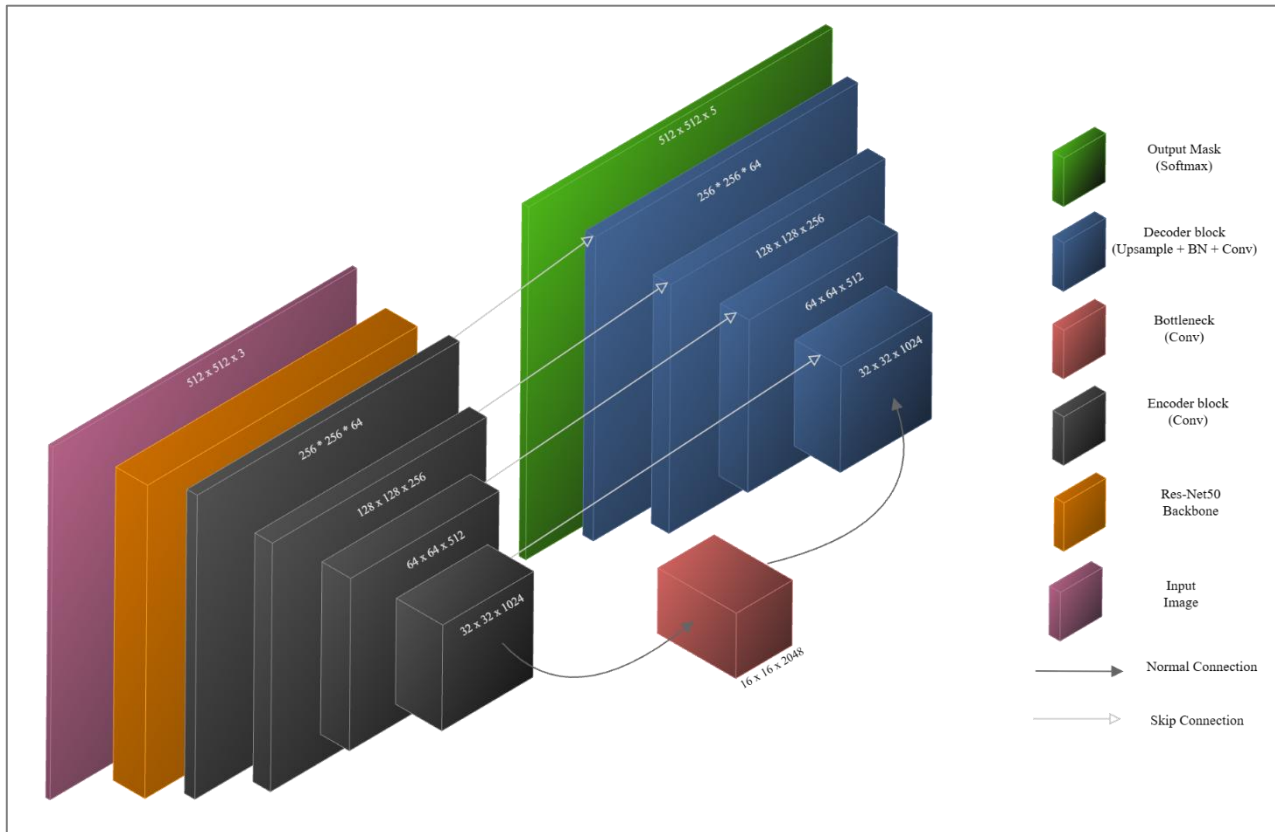


Fig. 3: U-Net Architecture Used in Implementation.

2. DeepLabV3: As shown in Fig. 4, we used a DeepLabV3+ architecture with an encoder-decoder architecture incorporating Atrous Spatial Pyramid Pooling (ASPP) that captures contextual information at all scales [17]. Instead of just ResNet-50, like with U-Net, we replaced the encoder with different pre-trained backbones on ImageNet weights. The Atrous Spatial Pyramid Pooling (ASPP) module applies parallel atrous convolutions with different dilation rates, and global pooling captures features at different scales. This allows for accurate segmentation of oil spills of all kinds of shapes and sizes. The decoder combines the output of ASPP and also computes low-level features to keep track of finer details. The combined feature maps are then upsampled to the resolution of the input image and passed through a SoftMax activation to generate the per-pixel labelled probability mask. DeepLabV3+ being able to model features at multiple levels and Precision localisation abilities is why it performs well in segmenting oil spills in SAR images.

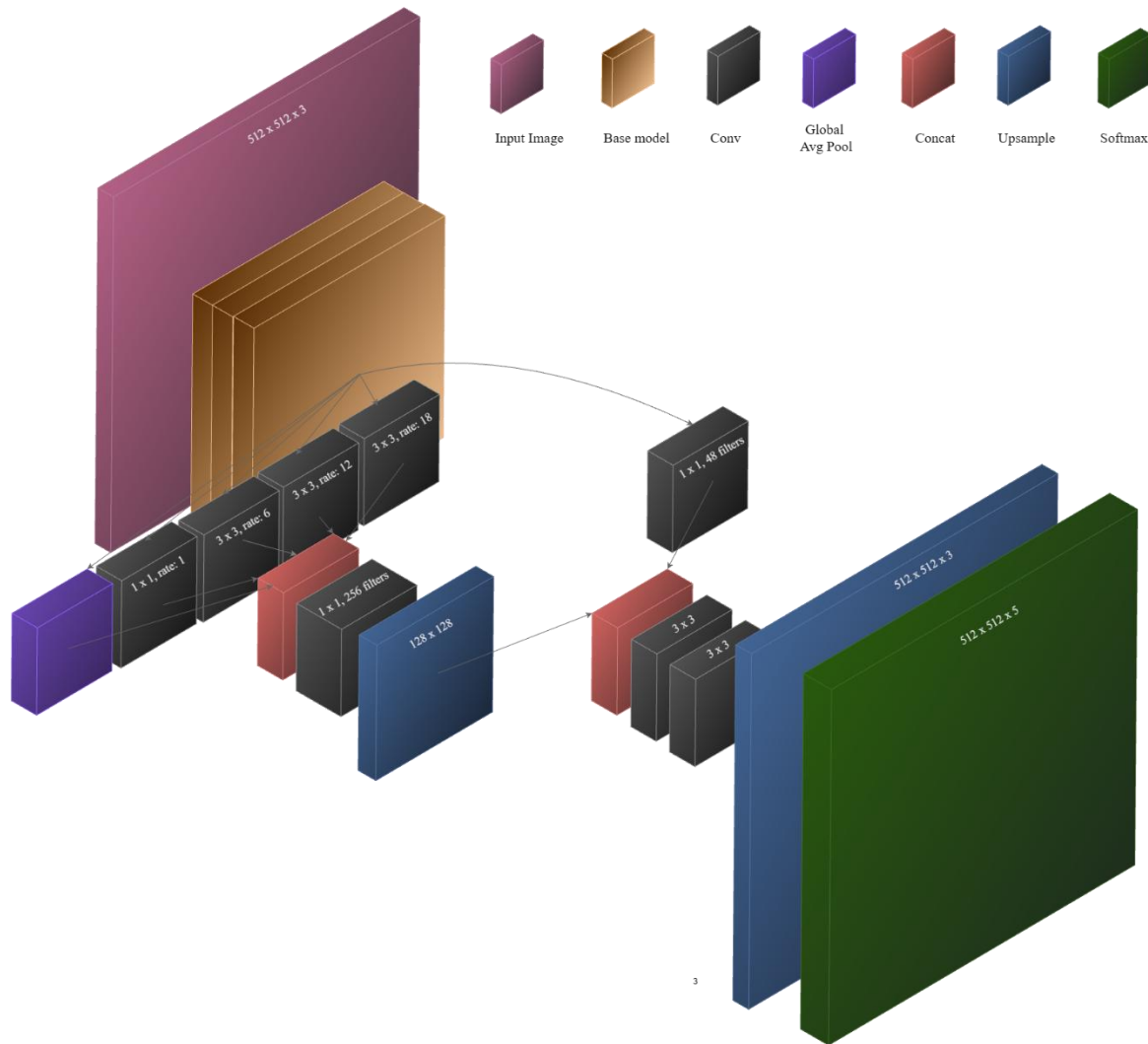


Fig. 4: DeepLabV3+ Architecture used in Implementation.

1 Loss Functions

2 To select a loss function is vital for squeezing model performance with problem statements having class imbalance and
3 accurate segmentation. In this investigation, we used four loss functions, as listed in Fig. 5: Categorical Cross-Entropy,
4 Dice Loss, Focal Loss, and Tversky Loss. From enhanced boundary identification to working with imbalanced datasets,
5 each of functions demonstrate different element in their learning process. Analysis and testing were carried out because
6 they were relevant to our use case.

7 1. Categorical Cross-Entropy (CE): Categorical cross-entropy measures the difference between the probability predicted
8 for each class and the actual true labels in a way that penalizes assigning higher probabilities to the incorrect classes. It
9 acts as a baseline loss, treating all classes equally. However, on its own, it isn't able to handle class imbalance.

$$L_{CE} = - \sum_{c=1}^C y_c \log(\hat{y}_c)$$

11 Where C is the number of classes

12 2. Dice Loss: Dice loss aims to improve pixel-wise overlap between prediction and actual labels. It is less sensitive to
13 class imbalance, better at boundary localisation, and therefore better at segmenting oil, which can be smaller in size as
14 well.

$$L_{Dice} = 1 - \frac{2 \sum y_i \hat{y}_i + \varepsilon}{\sum y_i + \sum \hat{y}_i + \varepsilon}$$

16 Where ε is a small constant added for numerical stability.

17 3. Focal Loss: Focal loss builds on categorical cross-entropy by using weighting. Dominant classes are down-weighted,
18 and the important classes that need to be focused on are optimised while training. Class-weighted focal loss counters
19 class imbalance and directs focus towards the oil-spill class.

$$L_{Focal} = - \sum_{c=1}^C w_c (1 - \hat{y}_c)^\gamma y_c \log(\hat{y}_c)$$

21 Where w_c is class specific weight. γ is the focusing parameter.

22 4. Tversky Loss: Tversky loss is a generalised version of Dice loss, where it asymmetrically penalizes false positives
23 and false negatives using arbitrary coefficients. It allows for more granular and intentional control over what kind of
24 errors are more acceptable comparatively.

$$L_{Tversky} = 1 - \frac{TP + \varepsilon}{TP + \alpha \cdot FN + \beta \cdot FP + \varepsilon}$$

26 Where α is the weight that controls the penalty for false negatives. β is the weight that controls the penalty for false
27 positives.

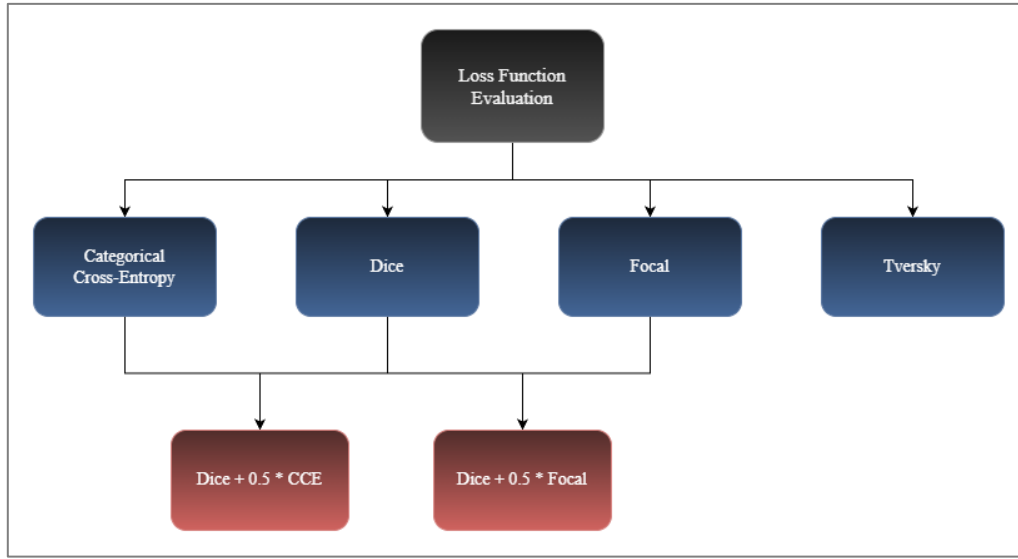


Fig. 5: Choices Tested as Loss Functions.

Evaluation Metrics

In this study, we are performing multi-class segmentation, but the primary target is improving performance over the oil-spill class. As a result, all the metrics mentioned from this point forward are oil-spill specific except for Mean Intersection over Union. Here, the metrics are computed using a one-vs-all methodology, where oil-spill class pixels are considered positives, and the rest of the classes are considered negatives.

1. Precision: Precision measures the ratio of correct positive predictions made out of all positive predictions made:

$$Precision = \frac{TP}{TP + FP}$$

2. Recall: Recall measures the ratio of correct positive predictions made out of actual positive values:

$$Recall = \frac{TP}{TP + FN}$$

3. F1-score: It is defined as the harmonic mean of Precision and Recall:

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

4. Intersection over Union (IoU): It measures spatial overlap between predicted and ground truth regions. It is measured across one class:

$$IoU = \frac{TP}{TP + FP + FN}$$

5. Mean Intersection over Union (mIoU): Mean IoU is the arithmetic mean of IoU over all the classes.

$$mIoU = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FP_c + FN_c}$$

Most pixels belong to the sea surface class, creating the class imbalance. In such cases, a model that simply predicts “sea” for all pixels will achieve high accuracy. Even though it doesn’t detect any oil-spills which is the primary objective of the model. This makes accuracy a weak indicator of model performance.

Model Evaluation

From Fig. 6, for choosing the optimal architecture, U-Net and DeepLabV3+ were considered. Both architectures were trained using the same training conditions that included a ResNet-50 backbone with pretrained ImageNet weights as the encoder, a joint loss function combining dice loss and focal loss, a constant learning rate of 0.0001 (1e-4), and Adam optimizer trained over 10 epochs. Class weights used for dice loss throughout this study were unchanged: they were arbitrarily assigned based on the number of pixels as follows: {0.05, 1.0, 3.0, 5.0, 3.0}.

$$L_{Joint} = L_{Dice} + 0.5 \cdot L_{Focal}$$

As Fig. 7 suggests, the U-Net achieved a higher Oil Recall and also generalized across classes better with a higher Mean IoU. However, DeepLabV3+ produced more conservative predictions, attaining better Precision, F1-score, and IoU on the oil class. Thus, suggesting a more balanced trade-off between actual detection and suppressing false alarms. These false alarms can be caused due to look-alikes in the form of aquatic vegetation, algae, and strong currents, which need to be distinguishable from actual spills. The higher Mean IoU on U-Net was mostly attributed to its better performance on the dominant sea-surface class. Given the study’s focus on oil spill detection, DeepLabV3+ made sense to be chosen as the preferred architecture. From this point onwards, the backbone selection will be performed on the DeepLabV3+ architecture.

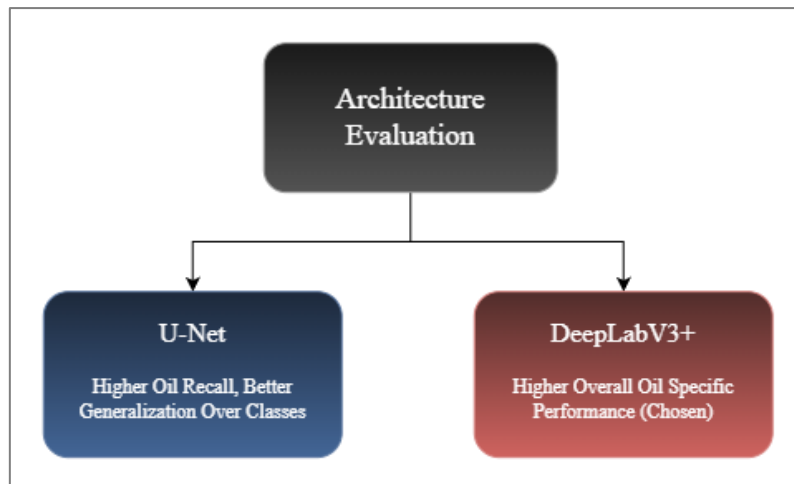


Fig. 6: Choices Tested as Architectures.

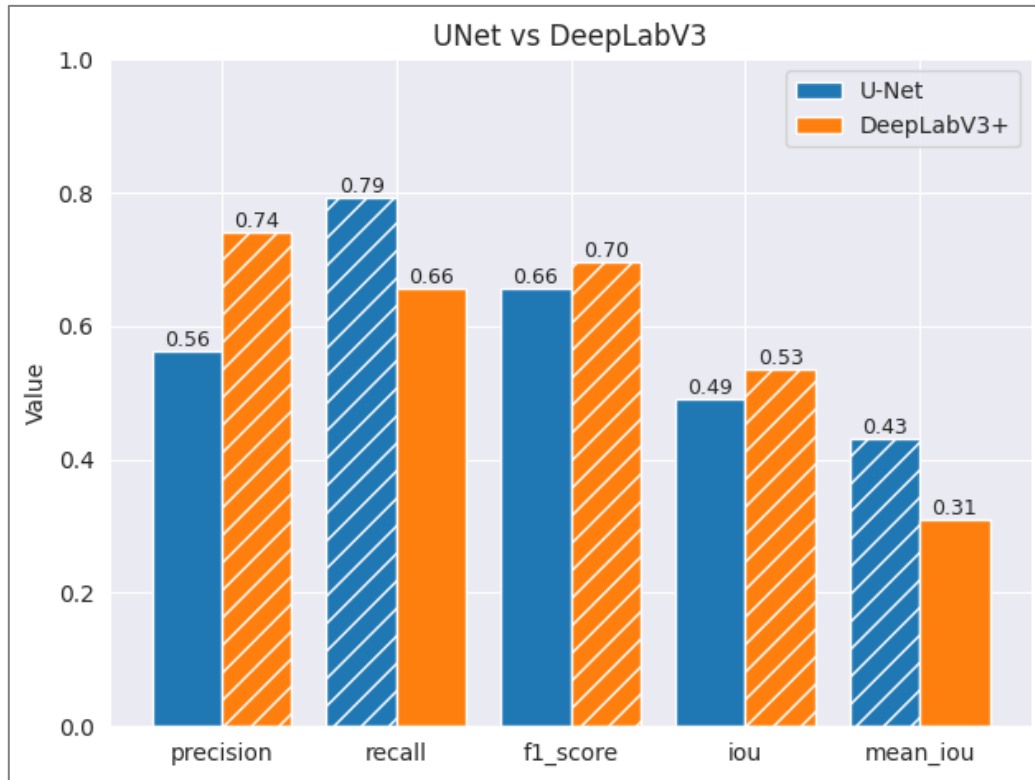


Fig. 7: Metrics Calculated Using Baseline U-Net and DeepLabV3+ Models.

After finalising DeepLabV3+ as the base architecture, from Fig. 8, different backbone networks for the encoder part were tested, namely ResNet-50, EfficientNetV2S, and Xception. As Fig. 9 suggests, they were trained under the same training conditions from architecture evaluation, only with a changed number of epochs and the backbone itself. Each model was trained up to 30 epochs, and performance was assessed at 10, 20, and 30 epochs to analyze convergence and ceiling performance. Xception is a variant of the Inception network, which replaces the standard convolutional blocks with depthwise separable convolutions. This decouples spatial and channel-based feature learning, making it efficient [18]. From the training curves, it is very evident that Xception was by far the best-performing model.

ResNet architecture introduced skip connections that prevented vanishing gradients and incorporated feature reuse [19]. ResNet-50 produced the highest loss and also plateaued with IoU and Mean IoU. EfficientNetV2S makes use of specialized MBConv and fused-MBConv blocks [20]. It also makes use of compound scaling, enabling faster learning. EfficientNetV2S also performed very well with comparable loss, IoU, and Mean IoU values to Xception throughout training. As Fig. 10 suggests, across all the metrics, Xception is clearly the best model. Xception has stable convergence behaviour and is the best performing on all fronts, and is therefore selected as the optimal backbone. Especially outperforming all other backbones on IoU and Mean IoU.

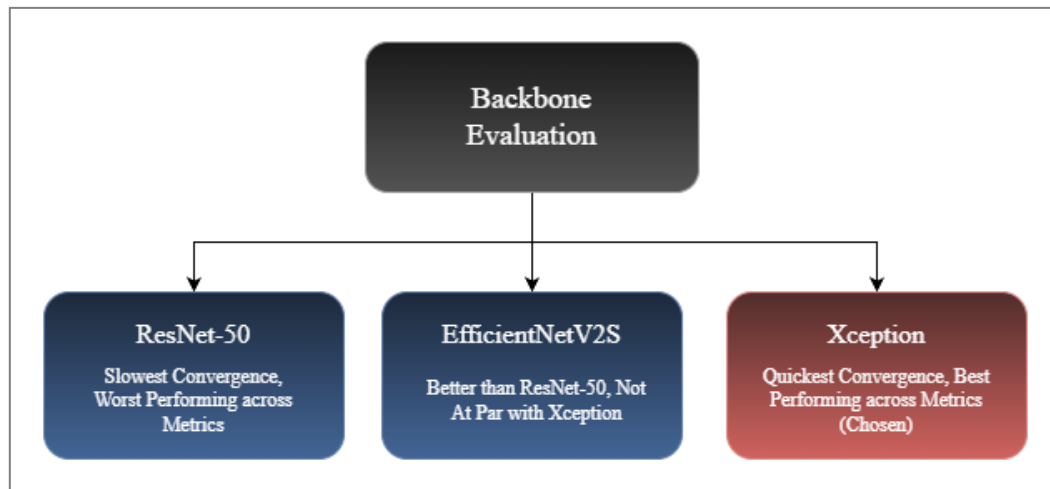


Fig. 8: Choices Tested as Encoder Backbones in DeepLabV3+

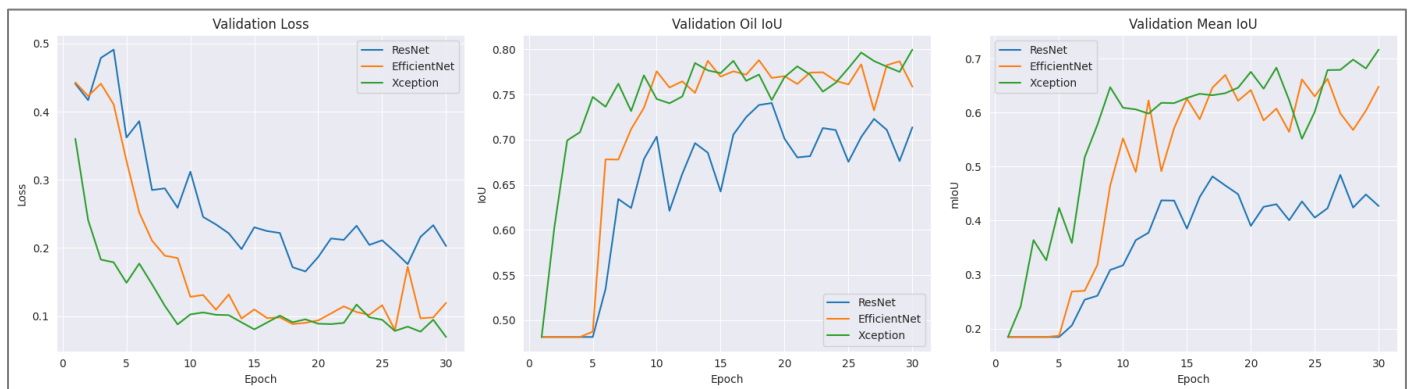


Fig. 9: Loss and Metrics Calculated while Training. Left: Validation Loss, Middle: Validation Oil IoU, Right: Validation Mean IoU.

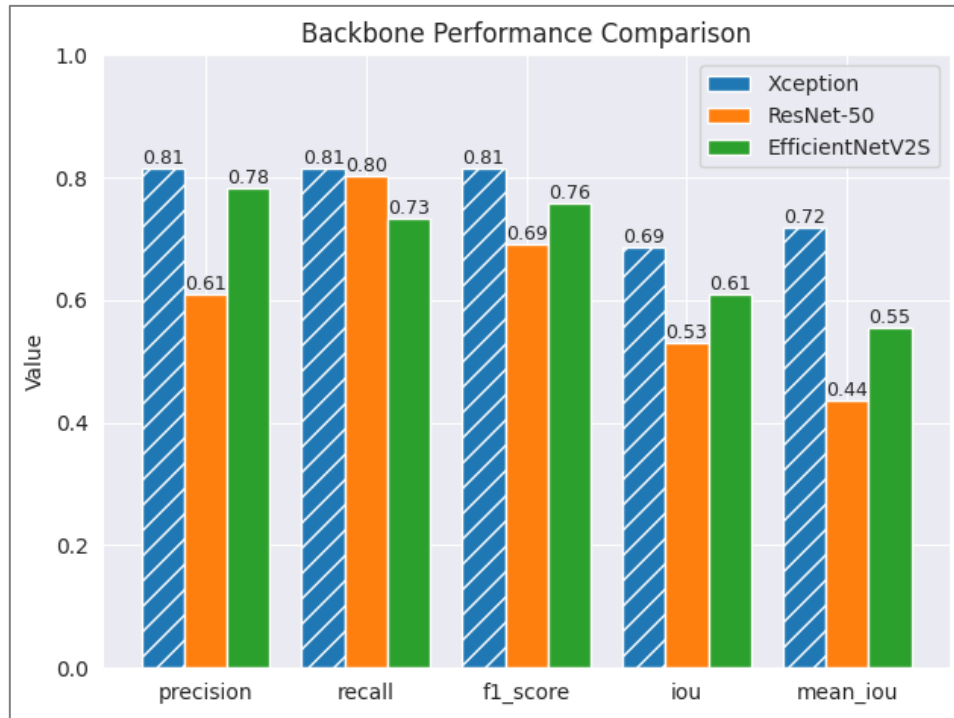


Fig. 10: Metrics Calculated on Testing Data with Xception, ResNet-50, and EfficientNetV2S Backbones.

All experiments performed till now involved the aforementioned basic Dice + Focal loss function. To select the best loss function for the model. Instead of the joint losses, single loss functions were tested. As shown in Fig. 11, Categorical Cross-Entropy, Dice, Focal, and Tversky Loss were first independently trained. Original Dice + Focal loss was also trained under the same conditions as a baseline comparison. Each model was trained for 30 epochs with early stopping rounds for 10 epochs, and also a callback that lowered the learning rate if the loss was plateauing. Dice Loss and Categorical Cross-Entropy were clearly the best performing, they were the only ones able to predict at par with the joint loss. Tversky loss fixated on Oil IoU and wasn't able to generalize well to the other classes. Focal Loss did the exact opposite and was able to generalize well, but didn't do well on Oil IoU.

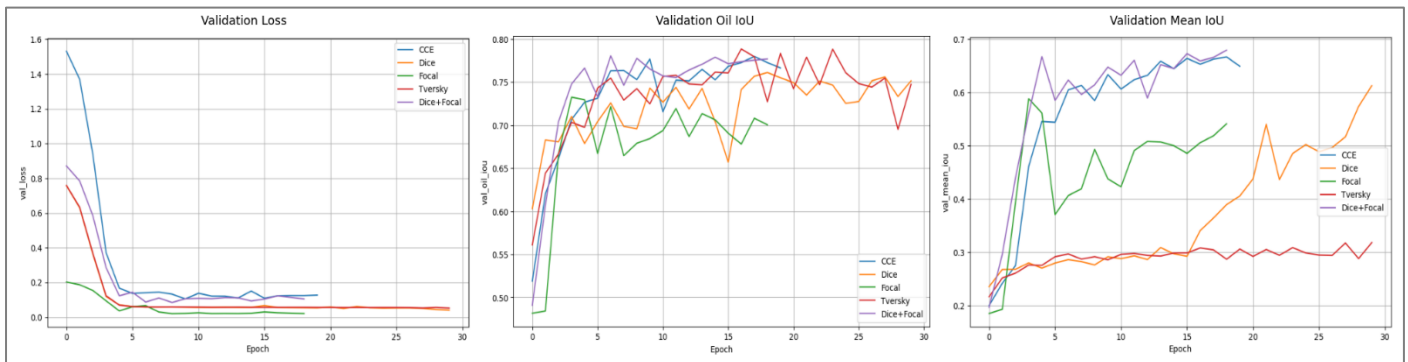


Fig. 11: Loss and Metrics Calculated with Different Losses. Left: Validation Loss, Middle: Validation Oil IoU, Right: Validation Mean IoU.

From Fig. 11, for evaluating model capabilities with joint loss, a new Dice + Categorical Cross-Entropy loss was formulated. As seen in Fig. 12, models with both losses were trained for 200 epochs each. For every chunk of 10 epochs, the best model was saved. For the first 100 epochs, the 10 best models were saved on the basis of the lowest loss. For the final 100 epochs, the 10 best models were saved on the basis of the lowest validation loss. In order to find the best model for each loss function, metrics were calculated using all 20 models on the test set. For both losses, the model had converged by 150 epochs. The best performing model for Dice + Focal loss was trained for about 190 epochs, and Dice + CCE for 170 epochs. The exact Dice + CCE loss equation used in implementation is as follows:

$$L_{Joint} = L_{Dice} + 0.5 \cdot L_{CCE}$$



Fig. 12: Loss and Metrics Calculated with Dice + Focal and Dice + CCE. Left: Training Loss, Middle: Training Oil IoU, Right: Training Mean IoU.

As shown in Fig. 12, the metrics calculated on the test set using the best models from each joint loss have been plotted. The newly formulated Dice + CCE loss outperforms the original loss marginally on all metrics. Apart from the target oil class, there are also other minority classes like Look-alike, Ship, and Land. In multi-class segmentation with class imbalance, accurate spatial recognition of these classes is important. From Fig. 13, we can observe class-wise IoU for all the classes. Dice + CCE loss outperformed the original loss on all classes except for Look-alike. Oil and Look-alike classes are the most similar; as a result, the IoU of Look-alike is lower for both models compared to the rest of the classes. The DeepLabV3+ model with Xception backbone and a Dice + Categorical Cross-Entropy Loss has emerged as the most accurate. As seen in Fig. 14, a prediction was made on a sample image and predicted mask with oil spill is seen.

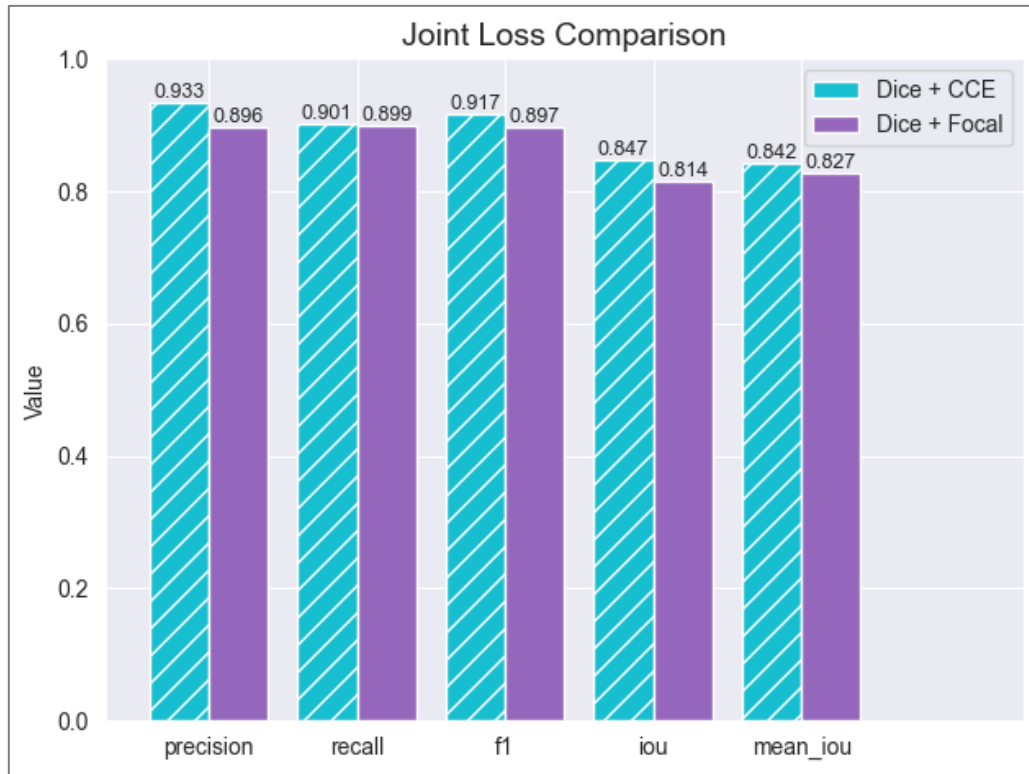
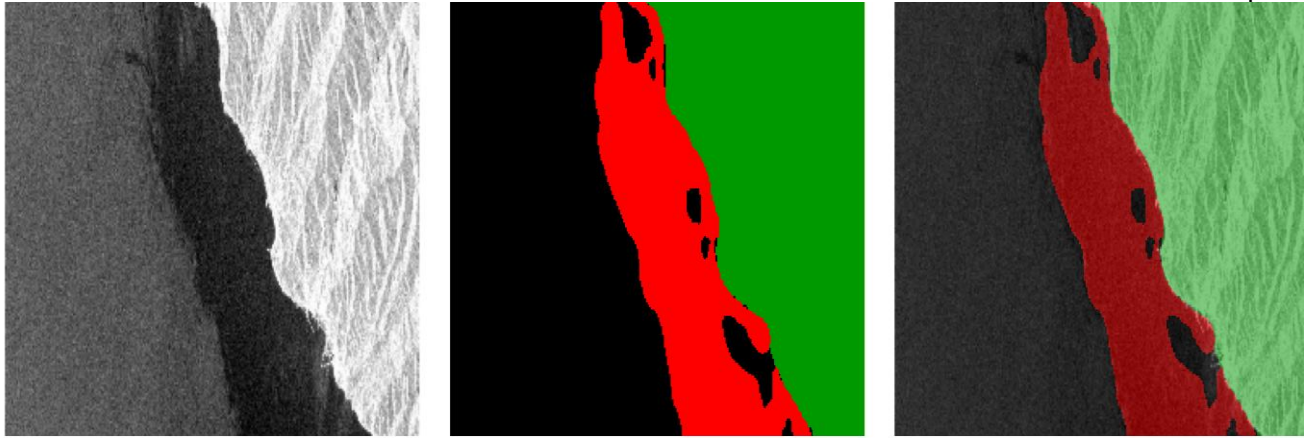


Fig. 12: Metrics Calculated on Testing Data with Dice + CCE Loss and Dice + Focal Loss Models.



Fig. 13: Class-wise IoU Comparison of Dice + CCE and Dice + Focal Loss Models.



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Fig. 14: Predictions Made on a sample 512 x 512 image patch by the final DeepLabV3+ model.

Left: Input 512 x 512 SAR Patch, Middle: Predicted Mask, Right: Predicted Mask Overlaid on Input Image.

Method validation

Table 1: Comparison of Our Work with Existing Studies

Ref & Year	Model Used	Dataset Used	Metrics - Value
[21] (2025)	DeepLabv3+	Sentinel-1 SAR (Suez Canal)	Oil mIoU: 0.7161, Mean IoU: 0.716
[22] (2025)	ResNet101 U-Net	Krestenitis SAR (Sentinel-1)	Oil IoU: 0.5806, Mean IoU: 0.6759
[23] (2025)	NOS-Net	Multi-class SAR	Oil IoU: 0.6127, Mean IoU: 0.7029
[24] (2024)	DeepLabv3+ improved	Sentinel-1 SAR (Gulf of Mexico)	Mean IoU: 0.8026, F1-score: 0.8866
[25] (2025)	Mask R-CNN	Sentinel-1 SAR	Oil IoU: 0.7480
[26] (2025)	LRA-UNet	Sentinel-1 SAR	Oil mIoU: 0.6736
[27] (2024)	MobileUNet variant	Sentinel-1 SAR	Oil IoU: 0.7585
Our Model	DeepLabV3+ (Xception, Dice + CCE Loss)	Sentinel-1 SAR	Oil IoU: 0.8467, Mean IoU: 0.8421, Oil Precision: 0.9334, Oil Recall: 0.9012, Oil F1-score: 0.9170

Table 1 compares the proposed approach with previously reported SAR-based oil spill detection methods. Similar comparisons are made in Fig. 15 and Fig. 16, where the IoU of existing research is compared with our model's IoU.

The DeepLabV3+ model from this work exceeds the existing IoU score, Oil F1-score, and Oil Recall; these improvements demonstrate the effectiveness of contextual feature learning and step-wise architecture, backbone, and loss evaluation strategies for SAR oil spill segmentation.

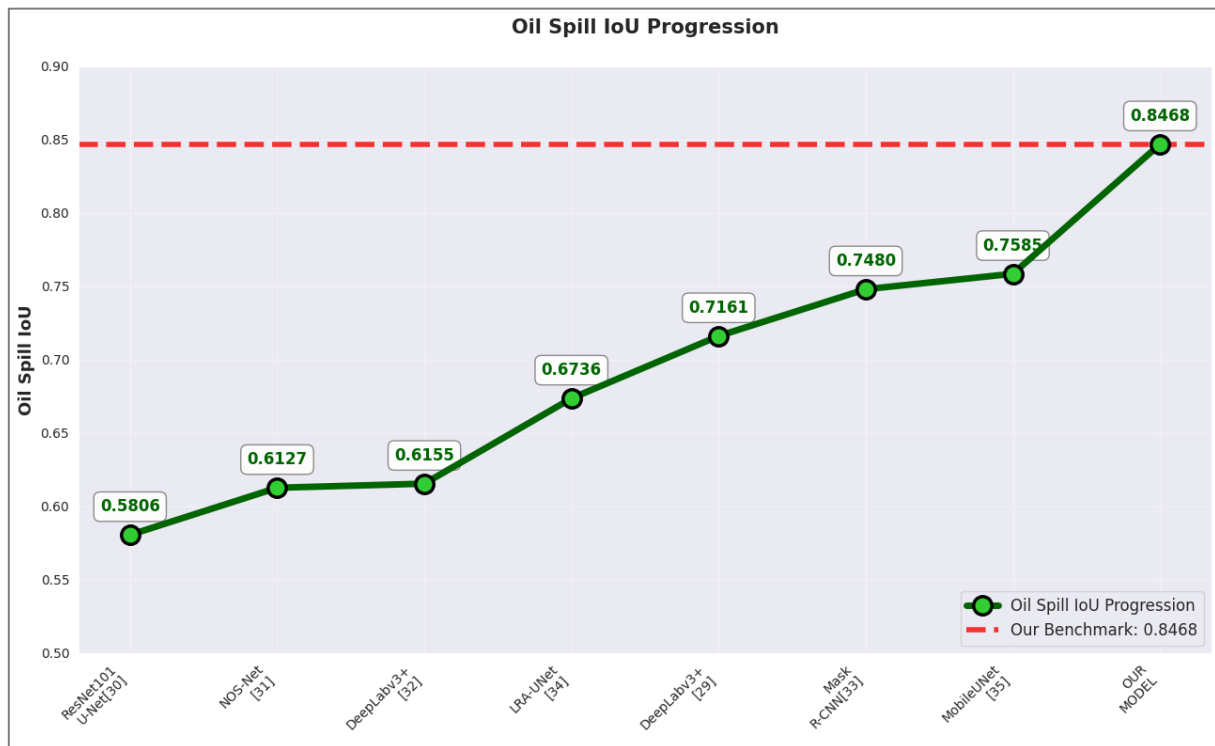


Fig. 15: Visualization of Oil Spill IoU Comparison Between Our Model and Existing Studies.

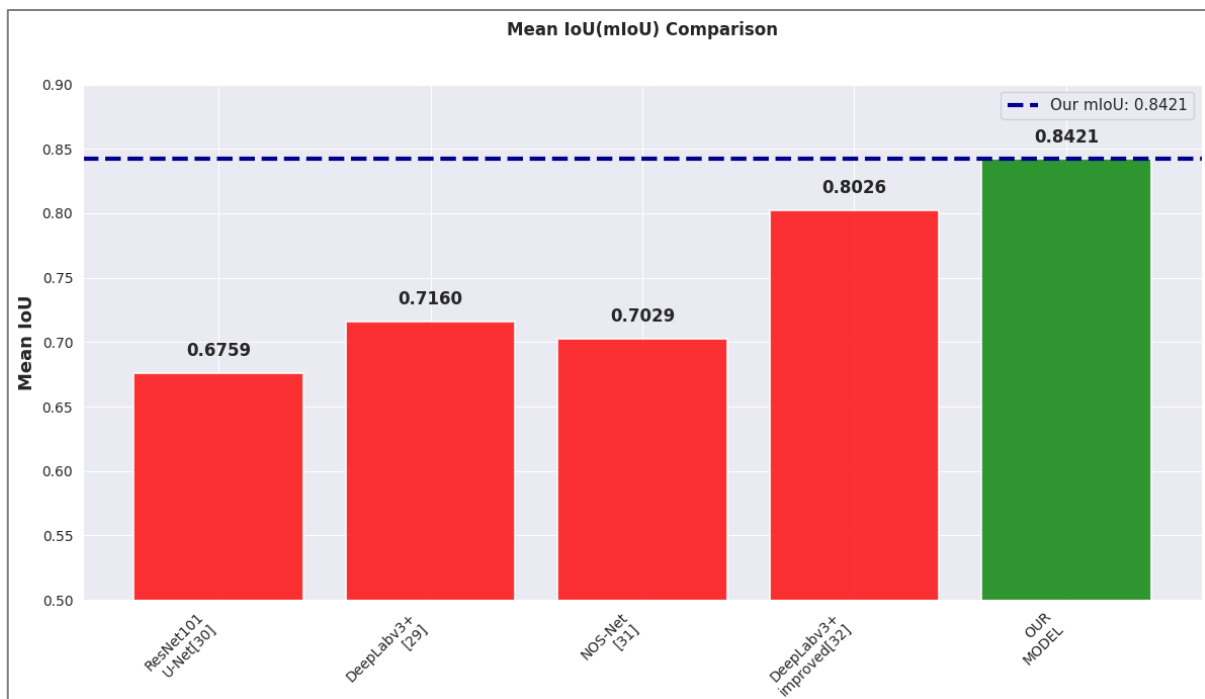


Fig. 16: Visualization of Mean IoU Comparison Between Our Model and Existing Studies.

Conclusion

This study presents an optimized deep learning model framework for the automated segmentation of oil spills in oceans using Sentinel-1 SAR imagery. By systematically evaluating the architectural configurations, backbones, and loss functions, it was found that a DeepLabV3+ model with an Xception backbone and a joint Dice + CCE loss function for evaluation provided the most robust performance. An integration of a patching strategy was enforced along with the training data formatting, improving the detection accuracy for small-scale spills, making it computationally efficient. Our approach successfully addresses the challenges of SAR imagery analysis, specifically class imbalance and the presence of “look-alikes” (e.g., algae or low-wind zones) by retaining patches of 1% oil-labelled. The model achieved an Oil-Spill IoU of 0.8467 and a Mean IoU of 0.8421, significantly outperforming the previous studies, such as LRA-UNet (0.6736) and improved DeepLabV3+ variants (0.8026). The model obtained an Oil Precision of 0.9334, an Oil Recall of 0.9012, and an Oil F1-score of 0.9170.

Future work will focus on integrating dual-polarization data (VV + VH) and exploring Vision Transformers (ViTs) to further enhance the global contextual understanding of maritime environments. These advancements contribute to more reliable, real-time satellite monitoring, essential for mitigating the environmental impact of maritime oil pollution.

Ethics statements

Not applicable

CRediT author statement

Bharti Khemani: Conceptualization, Methodology, Validation, Visualization, Investigation, Writing –review & editing, Supervision, Funding acquisition. **Jaya Gupta:** Conceptualization, Methodology, Validation, Visualization, Investigation, Writing –review & editing, Supervision, Funding acquisition. **Raj Nikam:** Conceptualization, Methodology, Validation, Data curation, Visualization, Writing –original draft, Investigation. **Prathamesh Mhatre:** Conceptualization, Methodology, Validation, Data curation, Investigation, Writing –original draft. **Tejashri Maske:** Conceptualization, Validation, Visualization, Investigation, Writing –original draft.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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